Suleyman Musayev

Senior Staff Engineer · Distributed Systems & Scale

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SUMMARY

Sole architect of a distributed authentication and identity platform at SSL.com, a top-10 global PKI vendor - serving tens of thousands of daily users and scaling the **core business 10×** in digital certificate issuance. Strong fundamentals in distributed systems, performance, and defense in depth. Open to senior and staff backend engineering roles at scale.

EXPERIENCE

Senior Staff Engineer · SSL.com

Aug 2023 - Present

Remote · top-10 global PKI vendor · sole architect of auth & identity platform

- **Sole architect** of a high-scale distributed authentication and identity platform serving tens of thousands of daily users; scaled the **core business 10×** across TLS, code signing, and identity certificate issuance.
- Defined the architecture for a distributed system of authentication services, API servers, web applications, and audit-logging pipelines; **led the redesign of the auth engine** for high-volume load and long-term scalability.
- Built a fully automated certificate issuance pipeline and complex API endpoints for certificate lifecycle management at scale.
- Implemented **OAuth 2.0, OIDC, SAML, and session-based** authentication with cross-domain flows and external identity provider integrations.
- Designed fraud and abuse defense at scale: bot detection handling **1,000+ fake signups/day**, insider-threat detection, behavioral analytics, IP intelligence, and device fingerprinting; **shut down an SMS pumping attack in 4 days**, closed account-takeover and second-factor-bypass paths before exploitation, and defeated persistent, sophisticated attackers at the application layer.

Software Engineer · Midstay

Aug 2022 - Jan 2024

Remote · early engineer, 5-person team, ~1k DAU consumer product

- Reduced backend latency by up to **5×** through query optimization, indexing, and caching strategy; improved user retention by **15%**.
- Shipped 4 major product features in 12 months; led incident response against advanced attackers. Final 5 months were a contract wind-down during transition to SSL.com.

SKILLS

LANGUAGES	Ruby, JavaScript, Python, SQL, C, C++
BACKEND	Ruby on Rails (4-8), REST API design, Sinatra, Sidekiq, background processing
DISTRIBUTED SYSTEMS	Service decomposition, message queues, audit pipelines, capacity planning, load balancing, query optimization, caching strategy
DATA	PostgreSQL, Redis, indexing strategy
AUTH & IDENTITY	OAuth 2.0, OpenID Connect (OIDC), SAML, JWT, Session-based, mTLS, SSO, Cross-domain auth
SECURITY	PKI, Certificate Lifecycle, Fraud Detection, Behavioral Analytics, Device Fingerprinting, MPIC, Threat Modeling
INFRASTRUCTURE	Docker, Linux, Nginx, AWS, GitHub Actions, CI/CD

OPEN SOURCE & LANGUAGES

Maintained gems: omniauth_oidc (OIDC for Microsoft Entra & Cloudflare), ip_geo_lookup, balloon_hashing, yescrypt. Contributor to Authlogic and openmpic.

Languages: English (Professional) · Russian (Native) · Turkmen (Native) · Turkish (Conversational)